Electromagnetic Core-Mantle Coupling in Mercury's Rotation

UCLouvain
Earth and Life Institute

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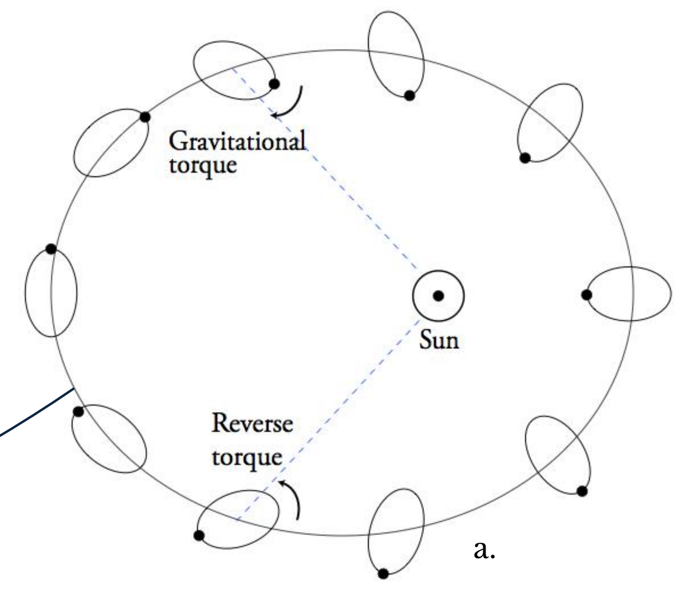
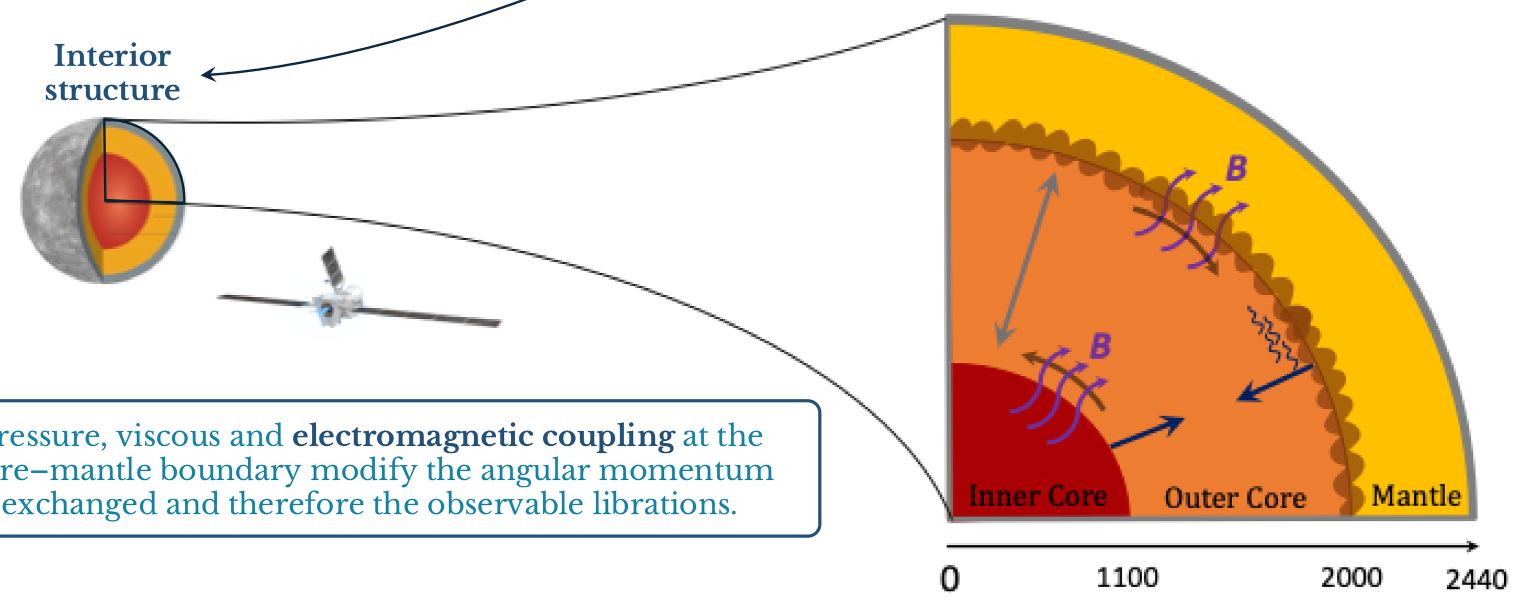
Alexandre Nuyt [1] , Jérémy Rekier [1,2] , Véronique Dehant [1]
[1] Earth and Life Institute, UCLouvain, Belgium [2] Royal Observatory of Belgium

Scientific motivation

Mercury is locked in a 3:2 spin–orbit resonance. Because of its eccentric orbit and triaxial shape, the solar gravitational torque varies periodically, producing **forced longitudinal librations.**

The amplitude and period of these librations depend on how angular momentum is exchanged between the **mantle** and the **fluid core**.





Pressure, viscous and **electromagnetic coupling** at the core–mantle boundary modify the angular momentum exchanged and therefore the observable librations.

Current models

Thin-wall approximation : Limited description of electromagnetic coupling.

If the **skin-depth**

$$\delta_m = \sqrt{\frac{2}{\mu\sigma\omega}} \leq d$$

is smaller than **the electrical conducting depth d** of the mantle, the approximation is not valid anymore.

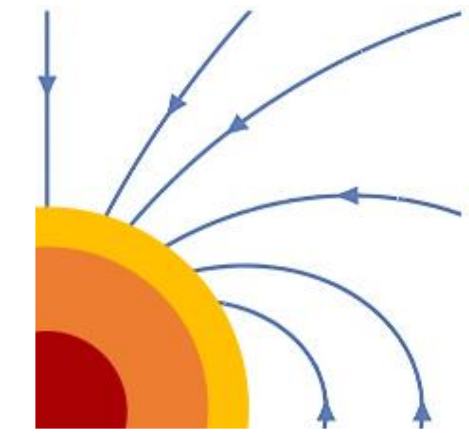
Objective

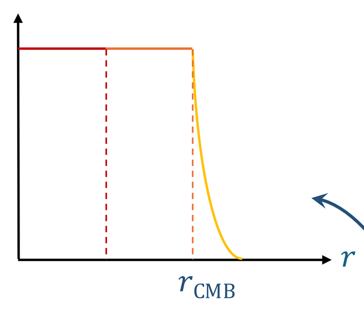
Three-layer model : Realistic **electromagnetic coupling** by solving **magnetic diffusion** throughout the mantle.

Scientific goal

Interpret future **BepiColombo** rotation measurements.

Methods





KORE (Rekier, J. & Triana, S. A. , 2024)
MHD in the fluid outer core
+
This work
Magnetic diffusion in the mantle
Variable conductivity profile $\sigma(r)$

Fully coupled 3-layer model

Inner core – Fluid outer core – Mantle

Induction equation under different forms

General Induction equation form

$$\frac{\partial \boldsymbol{B}}{\partial t} = \nabla \times (\boldsymbol{u} \times \boldsymbol{B}) - \nabla \times (\eta(r) \nabla \times \boldsymbol{B})$$

Inner core : no advection, η cst

$$\frac{\partial \boldsymbol{B}}{\partial t} = \eta \nabla^2 \boldsymbol{B} \longrightarrow \text{Sph. Bessel's functions}$$

Fluid core : η cst

$$\frac{\partial \boldsymbol{B}}{\partial t} = \nabla \times (\boldsymbol{u} \times \boldsymbol{B}) + \eta \nabla^2 \boldsymbol{B}$$

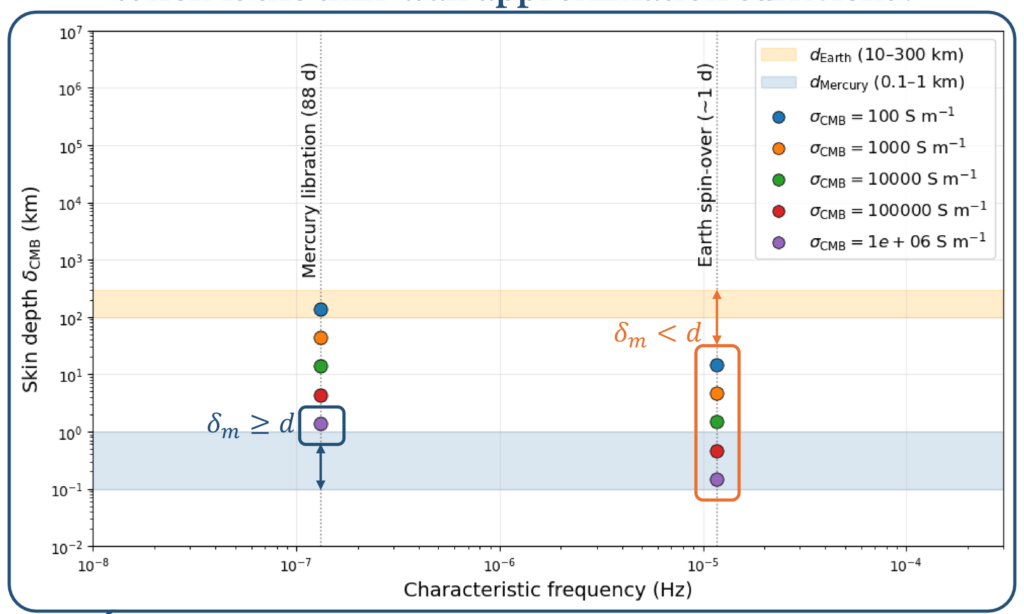
Mantle : no advetion

$$\frac{\partial \boldsymbol{B}}{\partial t} = -\nabla \times (\eta(r) \nabla \times \boldsymbol{B})$$

With a variable diffusivity/conductivity profile

$$\sigma(r) = \sigma_{CMB}\, e^{\alpha(r_{CMB} - r)} = \frac{1}{\mu_0 \eta(r)}$$

When is the thin-wall approximation sufficient ?



Development

Generalized Spherical Harmonics
Poloidal-Toroidal decomposition

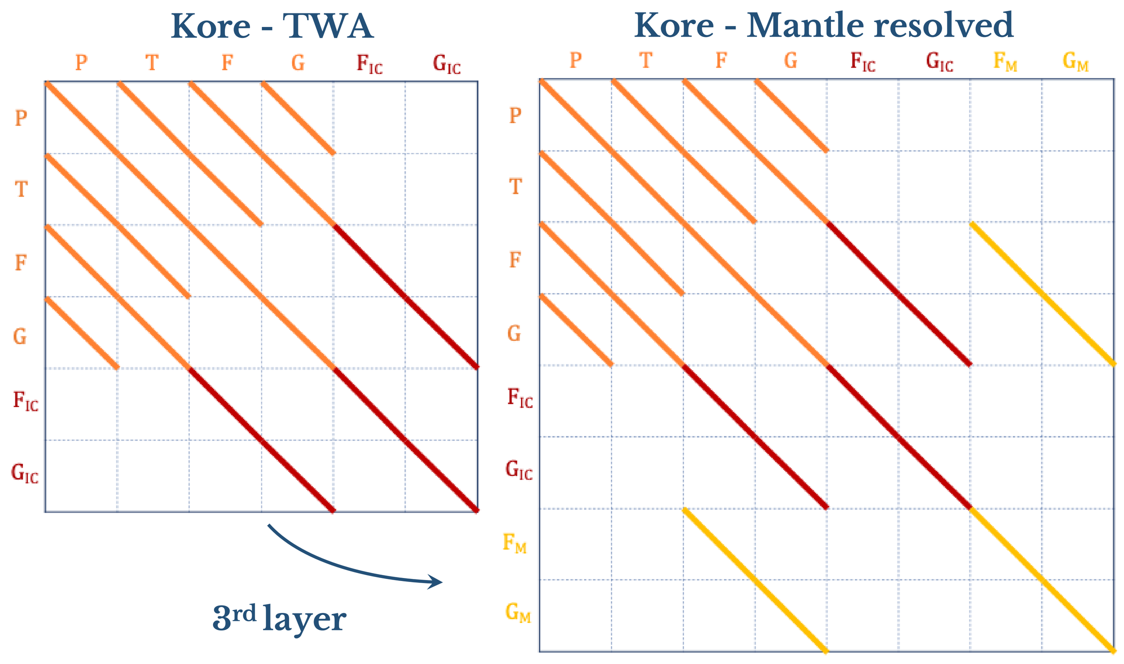
Spectral discretization

Ultraspherical Spectral Method
Chebychev Polynomial Basis
(Olver and Townsend, 2013)

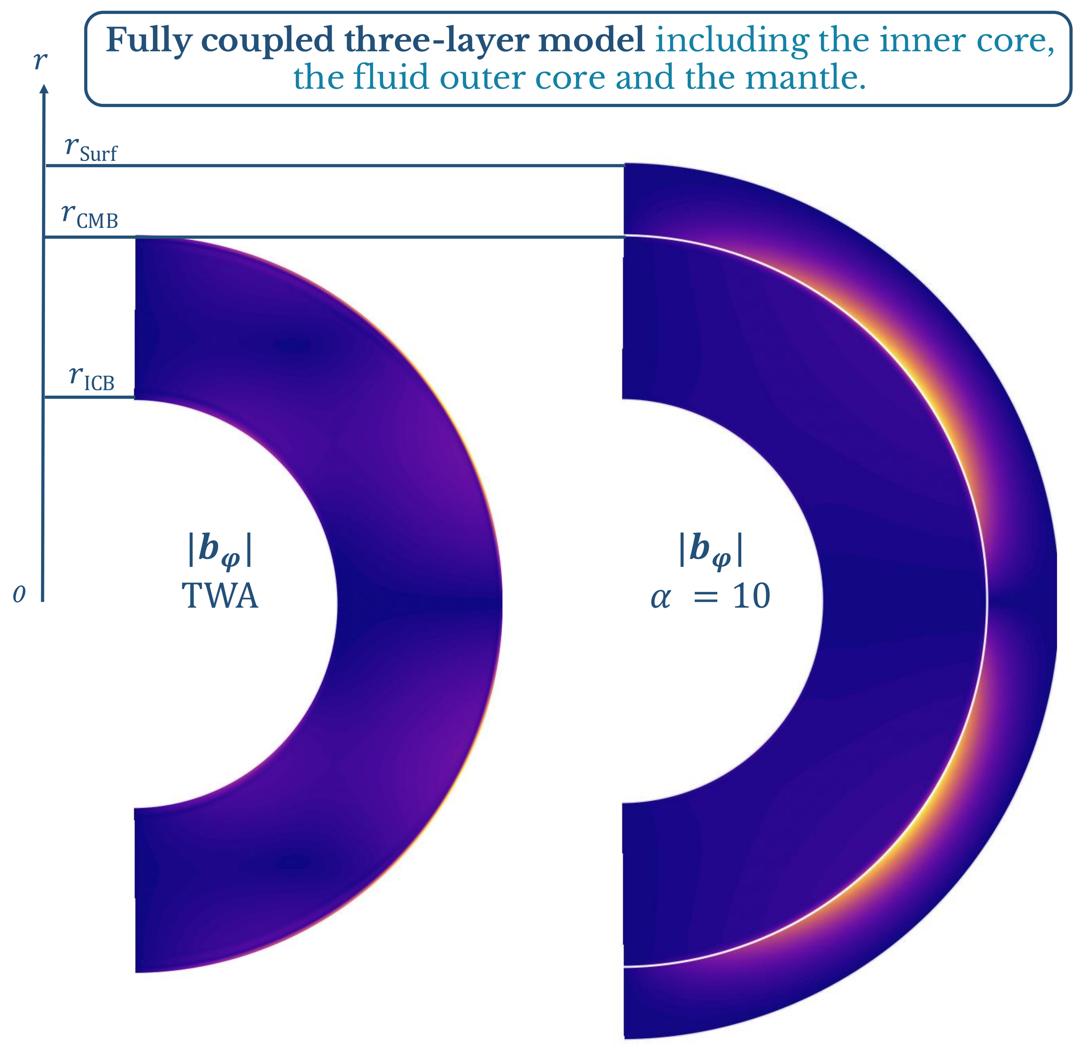
Assembly

Matching conditions for $\boldsymbol{b}$ field from the ICB to the surface
Coupled eigenvalue / forced problem

Results



Fully coupled three-layer model including the inner core, the fluid outer core and the mantle.



Why resolve the mantle ?

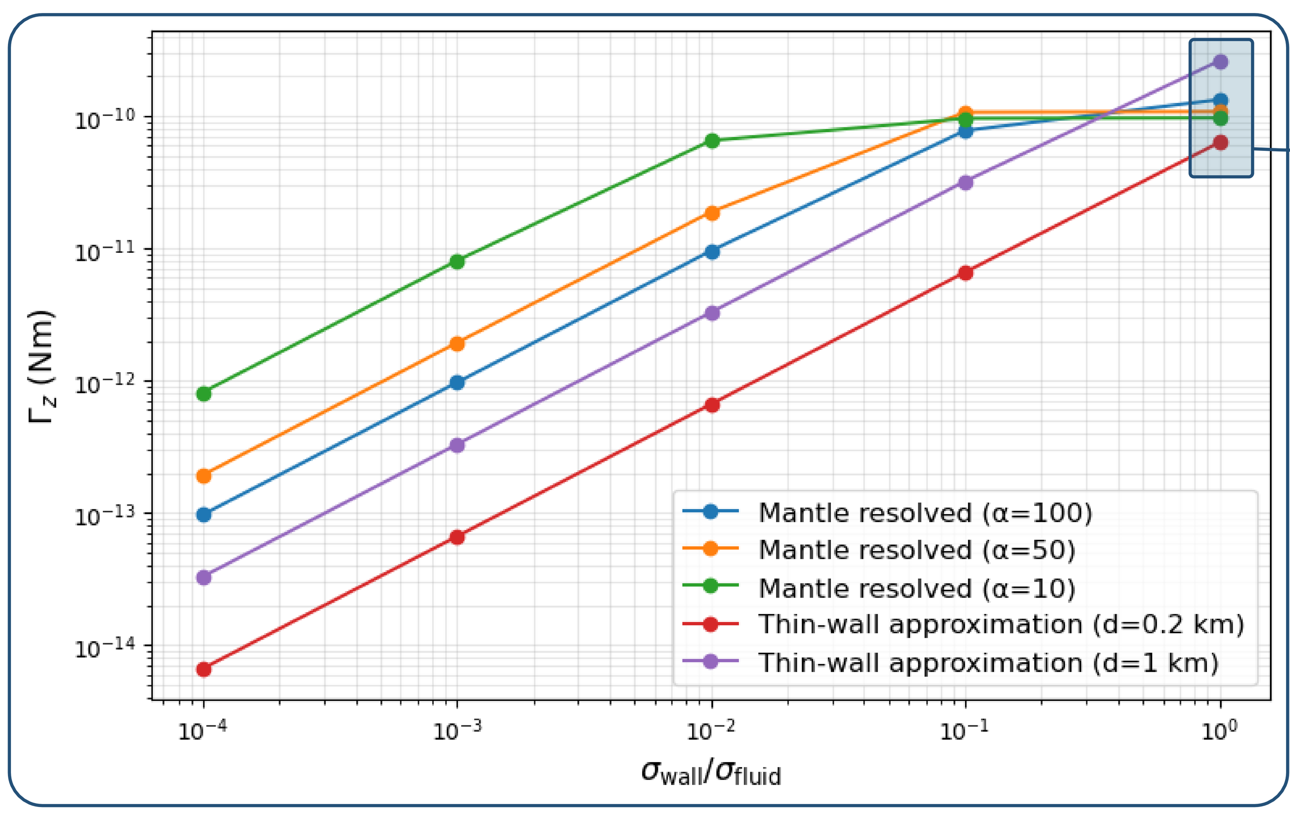
Mercury's 88-day libration lies in a regime of weak radial attenuation, providing **a natural test of the thin-wall approximation.**

Quantify the **validity range of the TWA**, which is close to the limit for high conductivity.

Systematically investigate **conductivity gradients** through realistic $\sigma(r)$ profiles.

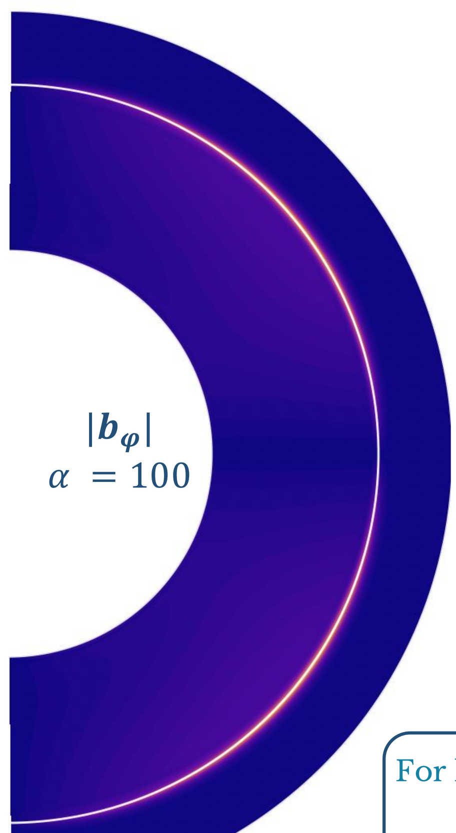
Provide a **general framework** applicable from Mercury's librations to Earth's rotational dynamics, imporant to **explore higher-frequency rotational modes**, where smaller skin depths make radial diffusion increasingly important.

Torque vs CMB conductivity

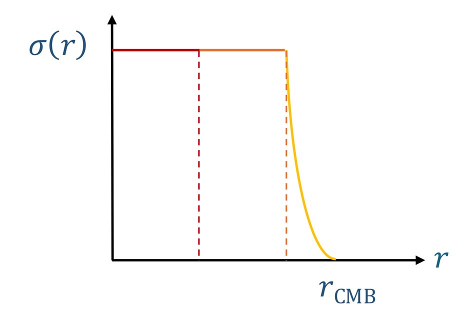


Parameter values used for the numerical computations

Ekman number, Ek $= 10^{-5}$; Forcing component, $m = 0$; Libration frequency, $\omega = 2/3$; Libration amplitude, $\epsilon = 1.9 \times 10^{-4}$.



Increasing α confines the magnetic field closer to the CMB.



For high **mantle conductance**,

$$G = \int_{r_{CMB}}^{r_{surf}} \sigma(r)\, dr$$

the different models tend to converge toward a similar torque.

This improved model

Quantifies how mantle conductivity modifies **electromagnetic core–mantle** coupling and will help assess its influence on Mercury's librations observed by **BepiColombo**.

Beyond Mercury

It will allow us to investigate **faster rotational modes**, such as **Earth's spin-over mode**, where magnetic diffusion becomes increasingly important.

Future Work

Will explore combine **electromagnetic and topographic coupling** at the CMB.

Kore

alexandre.nuyt@uclouvain.be

References

a. Figure adapted from Mathieu Dumberry's homepage (The librations of Mercury and its interior structure)

Rekier, J.& Triana, S. A. (2024). Kore as used in Rekier et al. 2024 [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.14058481

Olver, S., & Townsend, A. (2013). A Fast and Well-Conditioned Spectral Method. SIAM Review, 55(3), 462–489. https://doi.org/10.1137/120865458